Call Center Queuing Analysis

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Mode: Full Pipeline

1. Problem Formulation

Problem Type: Find
Domain: Queuing Theory

Universe

- Current Analytical: \{c, rho, p0, erlang_c, lq, wq_hours, wq_minutes, w_hours, w_minutes, l\}
- Simulation: \{wq_hours, wq_minutes, w_hours, avg_queue_length, customers_served\}

Variables

Symbol	Meaning	Type / Range
x	decision variable	\mathbb{{R}}

Structure

- **Arrivals**: Poisson process at lambda = 20 calls/hour
- **Service**: Exponential service time, mu = 6 calls/hour/agent
- **Current staff**: 4 agents (M/M/4 queue)
- **Questions**:
  1. What is the average wait time with 4 agents?
  2. How many agents are needed for < 2 minute average wait?
  3. What happens if call volume increases 50%?

Real-World Mapping

Real-World Concept	Mathematical Object
problem input	instance parameters

Constraints

(1) M/M/1 and M/M/c**: Markovian single-server and multi-server queues with Poisson arrivals

(2) Erlang-C formula**: Probability an arriving customer must wait in an M/M/c queue

(3) Utilization (rho)**: Server utilization rho = lambda / (c * mu); must be < 1 for stability

(4) Little's Law**: L = lambda * W (fundamental relationship between queue length and wait ti

(5) Warm-up period**: Discard initial transient observations in simulation before collecting stat

(6) Discrete-Event Simulation (DES)**: SimPy-based simulation to cross-validate analytical resu

Approach

Suggested approach: M/M/c Analytical + SimPy DES Verification

Complexity class:

Available tools: M/M/c Analytical + SimPy DES Verification

2. Solution

Answer:	Solution computed
Optimal:	No / Unknown
Feasible:	No
Algorithm:	M/M/c Analytical + SimPy DES Verification
Time:	2.8109s

Solution Details

Solution computed successfully.

Verification

✓	Utilization Below 1	Passed
✓	Littles Law Holds	Passed
✓	Wait Time Positive	Passed
✓	System Time Gt Service	Passed
✓	Simulation Matches Theory	Passed
✓	Staffing Reduces Wait	Passed
✓	Growth Needs More Staff	Passed
✓	Probability Bounds	Passed
✓	All Passed	Passed

3. Interpretation

The Question

- **Arrivals**: Poisson process at lambda = 20 calls/hour
- **Service**: Exponential service time, mu = 6 calls/hour/agent
- **Current staff**: 4 agents (M/M/4 queue)
- **Questions**:
  1.

The Answer

A feasible solution was found.

What This Means

- M/M/c analytical results: P0, Erlang-C P(wait), Lq, Wq, L, W for c=4 agents
- Little's Law check: L = lambda * W verified within tolerance
- Staffing analysis: Wq for c=1..10, minimum c for < 2 min average wait
- Growth scenario: Repeat analysis with lambda' = 30 calls/hour (50% increase)
- SimPy DES verification: 200,000-customer simulation (10,000 warm-up discarded), simulated Wq matches analytical Wq within 10%
- Staffing recommendation: Minimum agents for current and growth scenarios

Recommendations

1. Review the model assumptions to ensure real-world fidelity.

Limitations

- Model assumes deterministic parameters; real-world variability not captured.
- Solutio